

REPORT OF JAVA PROGRAMMING

Submitted To Mr. Anik

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE



By

Roll Number of Student: 25SCS1003005450

Name of Student: MANISH KUMAR LODHI

Batch: 2025-2029

Session : 2026-2027

CANDIDATE’S DECLARATION

I, Manish Kumar Lodhi, do hereby declare that the internship report titled “Java Programming” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references used in preparing this report have been acknowledged, and the report represents my own work based on the internship activities and project material provided for this submission.

I affirm that this report has not been submitted to any other institute for any degree or diploma requirement and that I shall be solely responsible for any kind of copyright violation in this regard.

Signature: Manish

Student Name: Manish Kumar Lodhi

Roll No.: 25SCS1003005450

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, constructive criticism, and friendly advice during this work. I am sincerely grateful to them for sharing their views and support on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature: Manish

Student Name: Manish Kumar Lodhi

Roll No.: 25SCS1003005450

INTERNSHIP COMPLETION CERTIFICATE



Internship completion certificate issued by CodeAlpha for the Java Programming Virtual Internship Program, dated 10th August 2026 to 10th September 2026. Student ID: CA/DF1/236218.

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4 PROJECT DESCRIPTION

4.1 Introduction

The Java Programming Internship at CodeAlpha provided practical exposure to core Java concepts and real-world application development. During the internship, the work focused on designing and developing projects using Java while strengthening understanding of Object-Oriented Programming (OOP), data structures, and problem-solving techniques.

As part of the internship, a Hotel Reservation System was developed as a console-based application. The system allows users to view available rooms, book rooms, cancel reservations, and manage booking records. The project provided practical experience in building structured, efficient, and maintainable software applications using Java.

Internship Details

Item	Details
Student Name	Manish Kumar Lodhi
Roll Number	25SCS1003005450
Internship Title	Java Programming
Organization	CodeAlpha
Duration	1 Month
Mode	Online
Type	Unpaid
Primary Technology	Java
Key Concepts	OOP, Collections Framework, Scanner Class
Batch	2026-27

4.2 Organization Profile

The internship presentation identifies CodeAlpha as the organization providing the Java Programming internship. The provided presentation does not contain a detailed company history, organizational structure, or business profile. Therefore, this section is limited to the organization information explicitly supported by the supplied material rather than adding unsupported claims.

The completion certificate supplied with this report confirms participation in the CodeAlpha Virtual Internship Program in Java Programming. It records the internship period as 10th August 2026 to 10th September 2026 and the date of issue as 10th September 2026. The certificate carries the student name Manish Kumar Lodhi and Student ID CA/DF1/236218.

4.3 Problem Statement

The project addresses the need for a simple console-based mechanism to manage hotel room reservations. The application needs to display available rooms, accept booking information, support reservation cancellation, and maintain booking records. The presentation describes these functions as the core operations implemented in the Hotel Reservation System.

The project also provides a practical programming problem through which Java classes, methods, conditional statements, loops, user input, and ArrayList-based data management can be applied together in one working application.

4.4 Project Objectives

- Develop a hotel reservation application using Java.
- Understand Object-Oriented Programming concepts.
- Learn how to manage data using ArrayList.
- Perform CRUD operations.
- Improve logical thinking and programming skills.
- Create a user-friendly menu-driven application.

These objectives are taken directly from the internship presentation and are aligned with the Hotel Reservation System developed during the internship.

4.5 Scope of the Project

The current scope covers a menu-driven, console-based Hotel Reservation System. Its demonstrated scope includes viewing available rooms, booking rooms, cancelling reservations, and managing booking records. The presentation also identifies possible expansion areas such as database connectivity, online payments, authentication, search and filtering, booking history, notifications, a graphical interface, and a web version.

- Room availability display.

- Room booking functionality.
- Booking cancellation.
- Booking-record management using ArrayList.
- Menu-driven user interaction.
- Testing and correction of logical errors.

The project is presented as a foundation that can be extended into a more complete hospitality-management application. The supplied material does not claim that database persistence, online payments, authentication, or a web interface are already implemented; these are listed as future enhancements.

4.6 Technologies and Tools Used

The presentation identifies Java as the programming language and lists the following concepts and development tools:

Category	Technologies / Concepts
Programming Language	Java
Programming Concepts	Object-Oriented Programming (OOP); Classes and Objects; Constructors; Methods; Loops; Conditional Statements
Data Handling	ArrayList / Collections Framework
Input Handling	Scanner Class
Development Environment	VS Code / IntelliJ IDEA / Eclipse
JDK	JDK 17+

4.7 System Architecture

The presentation does not provide a formal architecture diagram. Based strictly on the components and responsibilities described in the presentation, the project can be represented by the following logical structure:

Layer / Component	Role in the Project
User / Console Interface	Provides menu-driven interaction and accepts user input.
Application Logic	Processes room availability, booking, cancellation, and record-management operations.
Room Class	Represents room-related information used by the reservation system.
Booking Class	Represents booking/reservation information.

Layer / Component	Role in the Project
ArrayList / Collections	Maintains booking records and supports data management.
Testing & Debugging	Tests application functionality and fixes logical errors.

This is a logical interpretation of the project elements explicitly described in the presentation, not a claim that the original presentation contained a formal software-architecture diagram.

4.8 Methodology

The internship work followed a practical, project-oriented development approach. The presentation shows that the work began with Java fundamentals and OOP concepts and progressed toward implementation of a menu-driven Hotel Reservation System.

1. Understand the project requirements: room availability, booking, cancellation, and booking-record management.
2. Design the main entities using Java classes, including Room and Booking.
3. Implement methods, loops, conditional statements, and Scanner-based user input.
4. Use ArrayList to manage booking records.
5. Build the application around a menu-driven console interface.
6. Test the application functionality and identify logical errors.
7. Fix errors and improve the performance and readability of the code.

The methodology above is a structured presentation of the activities explicitly listed under the project roles, responsibilities, concepts used, and learning outcomes.

4.9 Expected Outcomes

The internship and project are expected to provide practical programming experience and a stronger understanding of Java-based application development. The presentation specifically identifies the following outcomes and benefits:

- Practical experience in Java programming.
- Application of theoretical Java concepts in a real project.
- Improved problem-solving ability and logical thinking.
- Understanding of the project development lifecycle.
- Improved debugging and testing skills.

- Experience with Object-Oriented Design and software development.
- Exposure to backend programming and database integration as a future direction.
- Foundation for advanced Java and enterprise application development.

During the internship, the student also learned Java fundamentals, OOP concepts, creation of multiple classes, ArrayList-based data management, user-input handling, menu-driven development, booking and cancellation logic, and clean, readable coding practices.

4.10 Certificates of Completion and Letter of Recommendation

The supplied Certificate of Completion confirms that Manish Kumar Lodhi successfully completed the CodeAlpha Virtual Internship Program in Java Programming from 10th August 2026 to 10th September 2026. The certificate was issued on 10th September 2026 and carries Student ID CA/DF1/236218.

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support@codealpha.tech**LinkedIn:**www.linkedin.com/company/codealpha**Mobile No:**

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CODEALPHA

YOUR DREAM OUR PASSION

Letter Of Recommendation

TO WHOM IT MAY CONCERN

This is to certify that Manish Kumar Lodhi has successfully completed the Virtual Internship Program at CodeAlpha for a duration of one month from 10th August 2026 to 10th September 2026 as a Java Programming Internship.

He/She exhibited performance in this role and made a valuable contribution to our organization during the internship period. He/She has excellent analytical skills and He/She quickly acquired new skills and adeptly adapted to emerging technologies, demonstrating a high level of productivity.

He/She consistently displayed a willingness to provide assistance and fostered strong collaboration with fellow team members. His/Her dedication and skills make him/her a valuable asset to any prospective employer, and CodeAlpha wholeheartedly recommend him/her for any future endeavor.

REGARDS,

CA/DF1/236218

STUDENT I'D

FOUNDER & CEO



10th September 2026

DATE

The supplied Letter of Recommendation from CodeAlpha confirms the successful completion of the one-month Java Programming internship from 10th August 2026 to 10th September 2026. It records the student's analytical skills, adaptability, productivity, collaboration, and recommendation for future opportunities. The LOR carries Student ID CA/DF1/236218 and is dated 10th September 2026.

5 BIBLIOGRAPHY / REFERENCES

8. Java Programming Internship Assessment-II presentation supplied with this report (14-slide source presentation).
9. CodeAlpha Certificate of Completion supplied with this report.
10. CodeAlpha Letter of Recommendation supplied with this report.

Internship report format/template supplied by the student for IILM University, Greater Noida, U.P.

The certificate and Letter of Recommendation details in this report are based on the documents supplied by the student.

FUTURE ENHANCEMENTS

The presentation identifies the following future enhancements for the Hotel Reservation System:

- Database connectivity using MySQL.
- User login and authentication.
- Admin dashboard.
- Online payment integration.
- Room pricing system.
- Search and filter rooms.
- Booking history.
- Email or SMS confirmation.
- GUI using Java Swing or JavaFX.
- Web version using Spring Boot.

These items are future directions described in the source presentation and should not be treated as features already implemented in the current console-based project.